Final Report

Jack Vranicar

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1 Abstract

Typically, when selecting a spring, one can know with relative certainty what the spring constant will be. This is because the spring coefficient for certain spring designs and materials is well studied and characterized. However, if it is desirable to create an assembly solely with 3D printed parts, one will have difficulty applying these formulas. This is because it is difficult to reliably print a helical pattern with typical consumer grade 3D printers. To address this, an 'S' shaped spring model is introduced, and its response characterized. Sensitivity analysis is performed on this model to reduce the parameter set, and the results compared against a reference spring. Uncertainty Quantification is performed, and it is determined that this model can be reduced to two parameters. Future work is necessary to improve the confidence of the model.

2 Motivation

2.1 Mechanical Background

For Mechanical Engineers, there are a number of considerations that must be made when designing a system, especially if it is a dynamic system. These dynamic systems have moving components and are subject to numerous external forces. To react to these external forces, there are a couple of schemes that are typically used. The first is passive control. This refers to a system that can react to disturbances appropriately, simply by virtue of its compliance and damping. There is also active control, which includes using actuators and sensors to effect the desired system response. Often, passive and active control schemes are used together to reap the benefits of both schemes. For the purposes of this analysis, only passive systems will be discussed.

When engineers are designing systems, it is typical to go through a number of iterations. This process helps make small changes to designs, compare iterations to each other and ending up with the best product. It is becoming increasingly popular to create these prototypes out of 3D printed materials as they allow for cheap, fast, and in-house manufacturing of all components. However, this approach is typically limited to static applications- parts that fasten to each other or do not require precise passive control considerations. One usually does not 3D print precise compliant parts due to the complexity of characterizing the spring's coefficients. This can prove to be a serious limitation in a few cases, including a scale-model of a passively controlled system, or a passively controlled system that is intended to remain 3D printed.

If a 3D printed spring could be precisely characterized, a number of benefits would become available to designers. The flexibility that printing one's own springs opens a wide variety of designs previously impossible with traditional methods. This can lead to innovative, novel solutions which can compensate for limitations of other designs. For this reason, the goal of this analysis is to determine a model which can reliably determine the spring constant and damping coefficient of a 3D printed spring.

2.2 Statistical Background

In order for a model to be useful, the parameters of the model must be identified. There are two strategies used to determine these parameters: frequentist and Bayesian approaches. Bayesian methods for parameter determination will be used in this discussion.

The Bayesian approach requires that one treats each parameter as a random variable, as opposed to the frequentist approach of treating each parameter as a fixed variable. This randomness is useful as it can account for uncertainties in production, such as measurement and manufacturing errors. Each of the parameters can then be defined as a Gaussian distribution with the mean value being the expected value of the parameter, and the variance representing the errors expected in producing the parameter. This is valuable for engineers as it allows for a better understanding of how these errors interact with each other and how confident they can be in the model prediction.

However, it is not necessary or even desirable to treat every parameter of a system as if it is random. The first reason to fix a model parameter is if its value is known with high certainty. These could be the results of measurements with high-precision measurement tools. Another reason deals with identifiability. If a parameter is identifiable, that means that the Bayesian inference can be used to determine an expected value. However, in cases when parameters are unidentifiable, the inference cannot yield a conclusion on the expected value of the parameter. This typically happens when two or more parameters are a function of each other. This is because the Markov Chain Monte Carlo (MCMC) method used in Bayesian analysis assumes that each of the parameters are independent and identically distributed (i.i.d.). If one parameter depends on the value of others, it is no longer i.i.d. Finally, fixing a parameter reduces the complexity of the modeling. Because the value for the parameter does not need to be sampled out of random distribution each time a point is evaluated, the MCMC requires less compute time.

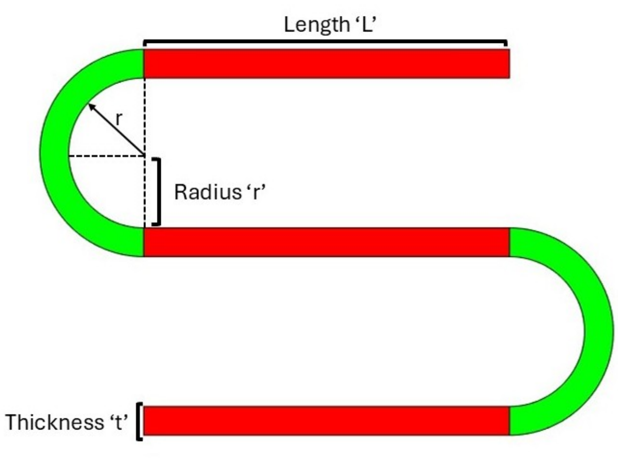
Two methods were used to determine which model parameters should be considered fixed: sensitivity analysis and MCMC. Sensitivity analysis is the process of quantifying how much a change in an input parameter changes the output parameters of the system. If a parameter is insensitive, it means that changing its value has little or no effect on the model's behavior. This is a means of determining that a parameter is unidentifiable as there is not definitive way of relating the parameter to the output of the system (see Section 3.2). As for MCMC, the chains of each parameter can be plotted against one another. If during this process a clear dependency is shown between variables, the parameter set must be reduced to ensure that the i.i.d. condition is satisfied (see Section 4).

2.3 Model Description

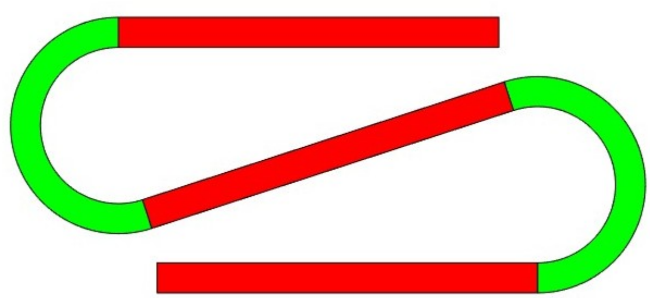
Four of the parameters of this system describe its geometry. Most of these parameters are displayed in Figure 1a. A parameter not shown in 1a is the Width w, which is the dimension into the page. This is a constant value for all of the components of the spring. Two more parameters, α and E, are included but do not relate to geometry. The parameter α relates to damping and is defined in Equation 4. E is the measure of the Young's modulus for the spring.

The set of parameters θ that will be considered random variables for uncertainty quantification are the aforementioned parameters:

$$\theta = \begin{bmatrix} L & r & t & w & \alpha & E \end{bmatrix}^T \tag{1}$$



(a) Spring Model: Uncompressed



(b) Spring Model: Compressed

Figure 1: Spring Model Compressed and Uncompressed

2.4 Model Dynamics

This system is modeled to behave as if it is a spring-mass-damper system. All of the mass is assumed to be concentrated at a point on the top of the spring. The damping is considered to be a linear function of velocity, and the stiffness is assumed to be a constant parameter based on physical and geometrical properties. Also, the system is assumed to have only one degree of freedom which is vertical displacement.

The stiffness of this system is determined using the results derived by Wang et al. In this analysis, the characteristics of a micro-electro-mechanical system (MEMS) are determined through simulation and Finite Element Analysis (FEA). This MEMS structure is similar to the spring structure proposed in this analysis. The result of this analysis is that a stiffness matrix can be formed based solely on the properties of the materials and the shape of the MEMS structure. Because this derivation is quite complex, it will not be included. However, the stiffness matrix will herein be referred to as $\vec{K}$. [1]

Because this model has only one degree of freedom, the effective spring constant k_s can be extracted from $\vec{K}$ as shown in Equations 2 and 3. In Equation 2, u represents a unit vector in the direction of the model's degree of freedom.

$$u = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^T \tag{2}$$

$$k_s = u^T \vec{K} u \tag{3}$$

The damping of this system was assumed to come only from drag. Typically, drag is a function of the square of velocity. This causes the model to be more complex due to nonlinearities. To reduce the complexity, a first order Taylor series approximation of the drag is taken. Because the velocity of the spring is centered about zero, the reference point for this approximation is taken at a speed of zero. This approximation simplifies to the first-order time derivative of drag. Furthermore, a scaling term α is included to account for any unmodeled properties of the spring that contribute to drag. The damping coefficient used for the spring is shown below in Equation 4. This coefficient is assumed to be constant regardless of the direction of the springs motion.

$$k_d = \alpha C_d \rho A \tag{4}$$

where ρ is the density of air, and A and C_d are the cross-sectional area and coefficient of drag for the top surface of the spring, respectively.

3 Analysis

3.1 Optimization of Random Parameters

To analyze the behavior of the model, a reference, or 'ideal,' spring-mass-damper system was created. Then, a genetic algorithm was used to best fit the parameters θ of the model to this ideal behavior. In both cases, a state-space model was utilized to predict the output of the springs. Because the springs are considered to be massless, the mass of the spring was applied as a constant input force to the system. The state vector x is defined as:

$$x = \begin{bmatrix} y \\ \dot{y} \end{bmatrix} \tag{5}$$

where y is the total height of the spring and $\dot{y}$ is the vertical velocity of the spring.

The following parameters A, B, C, and D were used to construct the state-space model:

$$A = \begin{bmatrix} 0 & 1 \\ -ks/m & -kd/m \end{bmatrix} \tag{6}$$

$$B = \begin{bmatrix} 0 & 1/m \end{bmatrix} \tag{7}$$

$$C = \begin{bmatrix} 1 & 0 \end{bmatrix} \tag{8}$$

$$D = \begin{bmatrix} 0 \end{bmatrix} \tag{9}$$

where ks is the spring constant, kd is the damping coefficient, and m is the mass for both the idealized and modeled springs.

3.1.1 Defining the Ideal Behavior

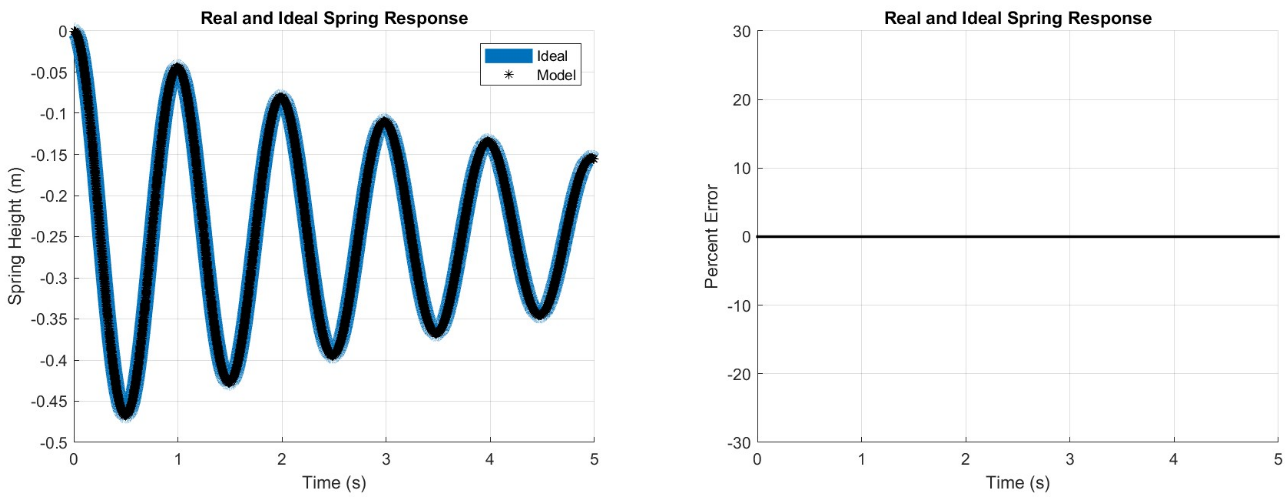
To define the ideal behavior, the values ks, kd and m were defined as 200, 2, and 10, respectively. Then, using the aforementioned state-space model, the ideal height response was defined as shown in Figure **??**.

3.1.2 Applying the Genetic Algorithm

A genetic algorithm was applied to determine the optimal set of parameters to match the model to the ideal response. The cost function used for this algorithm was the Root Sum of Squares (RSS) of the model's response and the ideal spring's response. The set of parameters to achieve this fit was found to be:

$$\theta_{opt} = \begin{bmatrix} 0.39 & 0.01 & 0.46 & 0.46 & 8.65 & 9.14E9 \end{bmatrix} \tag{10}$$

Figure 2a depicts the behavior of the ideal spring and the modeled spring, and Figure 2b shows the percent error between these two results.



(a) Ideal Spring Response

(b) Model Response

Figure 2: Ideal and Model Responses

3.2 Sensitivity Analysis

The parameter set θ_{opt} described in Equation 10 was used as the fixed point for sensitivity analysis using complex step methods.

There are other optimized sets of model parameters, however these combinations tend to create a spring which is not feasible to manufacture, namely because they become very tall and thin. For this reason, local sensitivity was considered to be appropriate to avoid entering these regions.

Complex step analysis works by approximating the derivative of the function with respect to a model parameter. The reason this method is used instead of finite differencing is that the step sizes can be much smaller, offering better accuracy. [2]

By applying complex step analysis over multiple test points for each model parameter, an approximation of the Jacobian of the model near the fixed point is defined as S. Each of these sensitivities is plotted against the test points that were used, as shown in Figure 3.

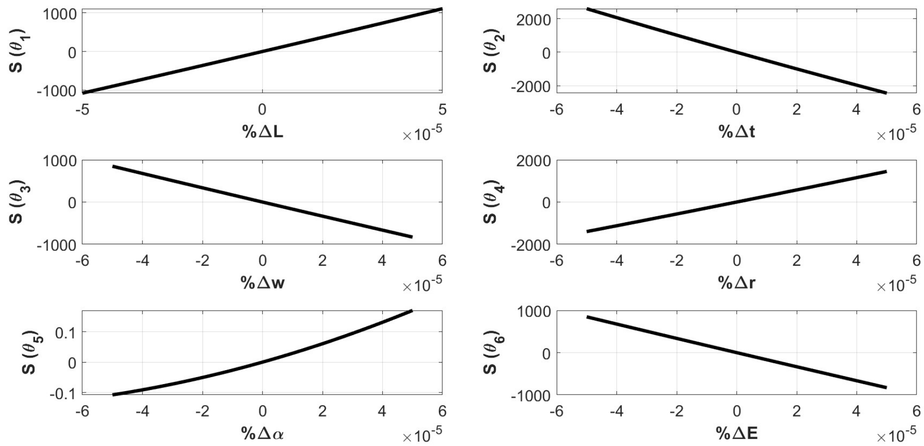


Figure 3: Local Sensitivities

From Figure 3, it can be concluded that θ_5 is insensitive and therefore is unidentifiable. This is due to the sensitivity of this parameter being multiple orders of magnitude smaller than the other parameters. For this reason, the parameter set, defined in Equation 1, can be reduced to:

$$\theta_{RED} = \begin{bmatrix} L & r & t & w & E \end{bmatrix}^T \tag{11}$$

4 Uncertainty Quantification

Uncertainty quantification for each of the parameters in the full set and the reduced set was conducted using MCMC methods.

4.1 Data Generation

Because no experiments were conducted, artificial data were created to run MCMC with. For a given ‘trial,’ random noise was added to each point of the ideal spring response. This was repeated for a total of 20 ‘experimental runs’ of the system. The random noise for each point was sampled from the distribution $(1/15) \sim R(0, 0.2)$. This data set is shown below in Figure 4.

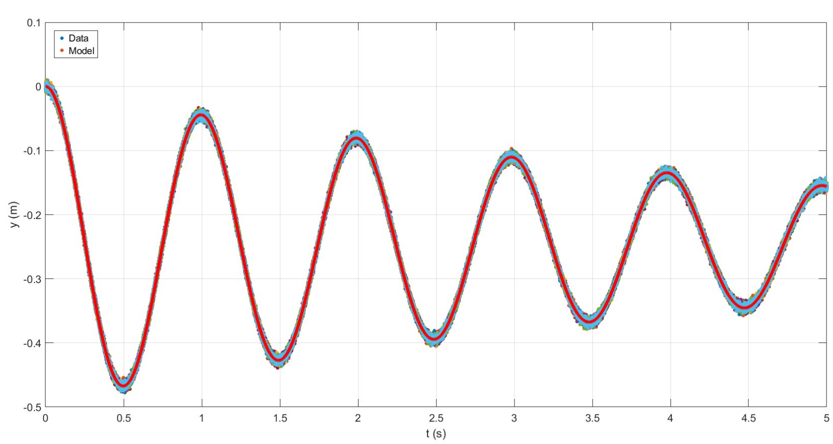
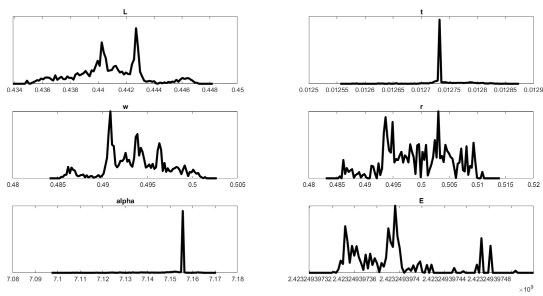


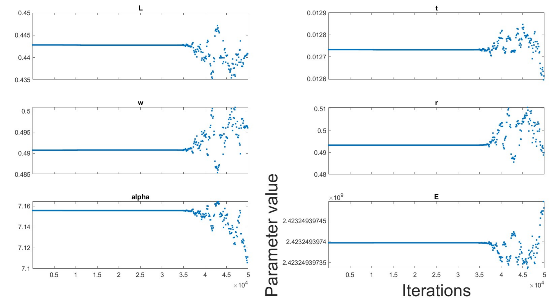
Figure 4: Simulated Data

4.2 Full Parameter Set θ

The results of the MCMC for the full parameter set is shown below. Figure 5a shows the posteriors for each of the parameters. The larger the y-value, the more likely that the parameter will minimize the cost function. Ideally, these peaks are sharp, showing high likelihood that the correct parameter value has been chosen. In this figure, there are not sharp peaks for each of the parameters. This indicates that the model has dependencies that have not been accounted for or that the chains are not ‘burned in.’ Burned in chains refer to chains which form a normal distribution about a singular value. In fact, Figure 5b shows that none of the parameters have reached a burned in region. This figure is a plot of the sampled value for the parameter versus the sampling iteration. The desired behavior is that the sampled points form a Gaussian distribution, which is not seen here. Reducing the parameter set, choosing different initial values, and increasing the sampling iterations of the MCMC algorithm will be used to improve these results.



(a) Parameter Chains Posteriors for θ



(b) Parameter Chains Iterations for θ

Figure 5: Ideal and Model Responses

The response of each parameter was plotted against the response of every other parameter, as shown in Figure 6. A number of dependencies between parameters are made apparent by this figure: w and L; and α and both L and w. The desired behavior for this plot is that there are no clear relationships between the parameters. Linear relationships, as are seen in the aforementioned parameter pairs, are an indication that the model is not i.i.d.

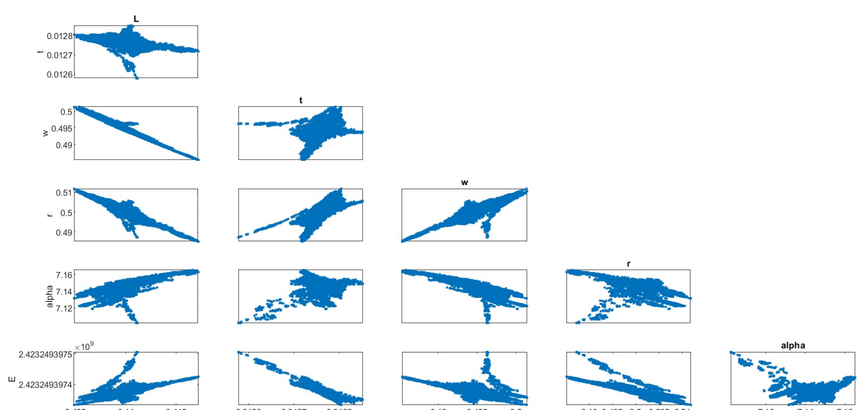


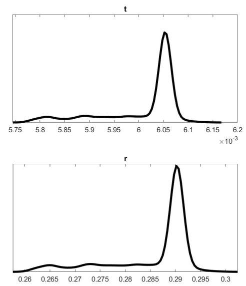
Figure 6: Parameter Pairs for θ

4.3 Reduced Parameter Set θ_{RED}

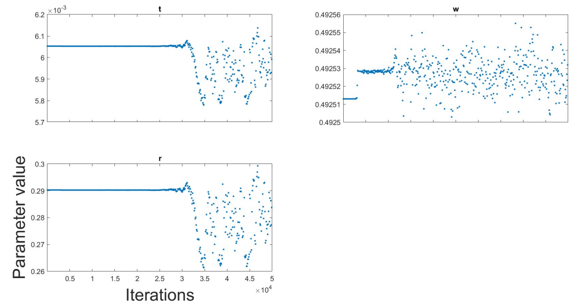
The parameter set θ is reduced to account for the results in Section 4.2, the sensitivity analysis results shown in Section 3.2, and iterative MCMC runs used to increase the number of identifiable parameters. Herein, θ_{RED} is defined by Equation 12.

$$\theta_{RED} = \begin{bmatrix} t & w & r & E \end{bmatrix} \tag{12}$$

The results of the MCMC for the reduced parameter set is shown below. Each of these plots are the same ones used in Section 4.2. Figure 7a shows that these parameters have begun to converge to singular values. These sharp peaks are indicative of higher confidence in the model parameters. Furthermore, in 7b, the values have begun to center around a single value and appear to form a Gaussian distribution. This is indicative of a burned in region, and further confirms the confidence of each parameter. The values for E begin to diverge as the iterations increased, which could be addressed by fixing the value of E or extending the number of sampling iterations.



(a) Parameter Chains Probabilities for θ_{RED}



(b) Parameter Chains Iterations for θ_{RED}

Figure 7: Ideal and Model Responses

Figure 8 below displays the parameter correlation for the parameters in θ_{RED}. Ideally, these parameters would not be showing a linear relationship with each other, as is seen with r and t. This is indicative of some fixed relationship in the parameters. Future work should include identifying this relationship, and reducing the parameter set where possible. By writing model parameters as a function of another model parameter, the number of degrees of freedom that the MCMC algorithm has is reduced. This strategy would yield non-correlated parameters.

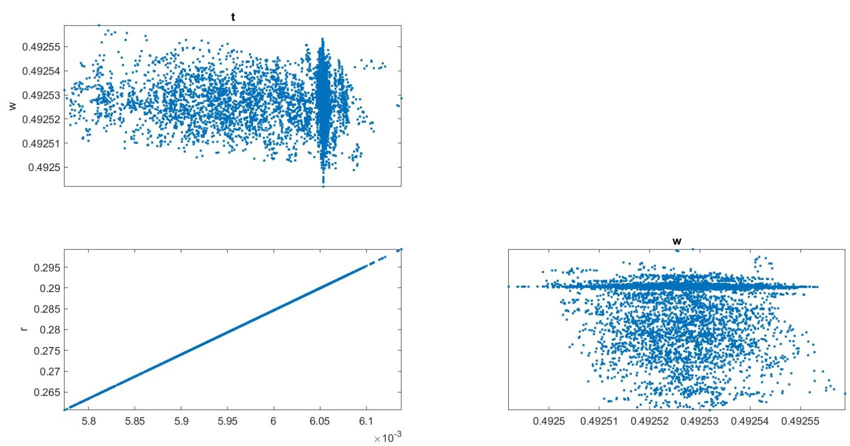
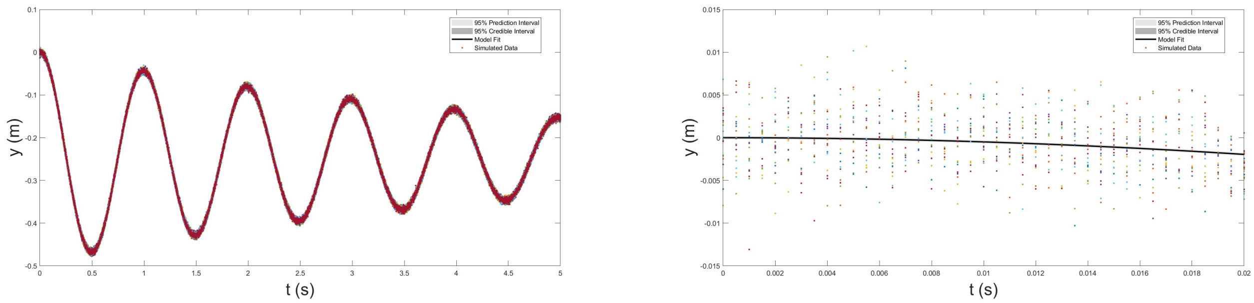


Figure 8: Parameter Correlation for θ_{RED}

Now that the parameter set is finalized, the uncertainty of the model can be quantified. This is shown below in Figure 9a. The 95% credible interval refers to the interval which has a 95% chance of containing the random parameter, and the 95% prediction interval refers to the interval which has a 95% chance of containing a future observation [2].

Because the 95% confidence interval does not contain 95% of the data, it is clear that there are abnormalities in the model. This is likely due to the violations of i.i.d. requirements discussed earlier. Future work should include identifying the sources of these errors and eliminating them.



(a) 95% Confidence Interval with θ_{RED}

(b) Zoomed In 95% Confidence Interval with θ_{RED}

Figure 9: Prediction and Confidence Intervals for θ_{RED}

5 Concluding Remarks

The MCMC methods were not able to satisfactorily converge on parameter values when the whole set of parameters θ were defined to be random, as demonstrated in Section 4.2. This required that the parameter set be reduced, yielding model parameters that converge appropriately. These results are shown in 4.3.

The most certain parameters was t and the least certain parameters were found to be $w, r, \& E$.

In Section 3.2, it is shown that the parameter α can be set as fixed as its sensitivity is far less than those of the other parameters. This is to be expected as the damping of the ideal system is relatively low, and α solely affects the damping k_d of the system. Otherwise, each parameter has a similar influence on the quantity of interest (RSS between modeled and ideal responses).

References

[1] N.F. Wang, Xianmin Zhang, and Zhiyuan Zhang. "Stiffness analysis of corrugated flexure beam using stiffness matrix method". In: *Journal of Mechanical Engineering SCience* 203-210 (2018), pp. 1989–1996.

[2] Ralph C. Smith. *Uncertainty Quantification: Theory, Implementation, and Applications*. siam, 2024.